MAIS PAC

IPhO 02.03: Dynamics: Energy, Momentum and Rotation · Worked Solutions

Dynamics: Energy, Momentum and Rotation: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam’s constants take precedence.

Solution index

1 2 3 4 5 6 7 8 9 10

Solution 1 — Work by a varying force. Return →

(a) $W = \int_0^3 6x^2\,dx = [2x^3]_0^3 = 54$ {J}.

(b) $K = (1)/(2)\ mv^2 = 54$ $v = \sqrt{(108/2)} = \sqrt{(54)} = 7.35$ {m,s^(-1)}.

(c) $P = Fv = (6\times9)(7.35) = 397$ {W}. **Check:** the force grows as x^2 and the speed grows as x^(3/2), so the power grows as x^(7/2) — steeply, as expected at large x.

Solution 2 — A block on a rough slope with a spring. Return →

(a) With total distance travelled s + d along the slope, the energy balance is

$$mg(s+d)\sin\theta - \mu\, mg(s+d)\cos\theta = (1)/(2)kd^2. \quad (1)$$

(b) This is a quadratic in d:

$$(1)/(2)kd^2 - mg(\sin\theta - \mu\cos\theta)d - mgs(\sin\theta - \mu\cos\theta) = 0, \quad (2)$$

whose positive root is the compression.

(c) For $\mu = 0$ it becomes $(1)/(2)kd^2 = mg(s+d)\sin\theta$, the standard spring–slope result. Dimensions: each term is an energy, so the equation is homogeneous. **Check:** if $\tan\theta < \mu$ the block never slides, and the quadratic's roots become unphysical — a sign that the model is being applied outside its range.

Solution 3 — Elastic and inelastic collisions from one formula. Return →

(a) For e=1 with u_2=0: v_1 = (m_1-m_2)/(m_1+m_2)u_1 = (1)/(3)(3.0) = 1.0 {m,s^(-1)} and v_2 = (2m_1)/(m_1+m_2)u_1 = (4)/(3)(3.0) = 4.0 {m,s^(-1)}. **Check:** momentum 2(1)+1(4) = 6 = 2(3) and kinetic energy 1 + 8 = 9 {J}, both conserved.

(b) e=0: common velocity = 6/3 = 2.0 {m,s^(-1)}.

(c) $\mu = (2)(1)/3 = 0.667$ {kg} and $(u_1-u_2)^2 = 9$. For e=0: $\Delta K = (1)/(2)(0.667)(9) = 3.0$ {J}. Directly: K_i = 9 {J}, K_f = (1)/(2)(3)(4) = 6 {J}, difference 3 {J} — matched. For e = 0.5: $\Delta K = 3.0(1-0.25) = 2.25$ {J}. **Check:** the energy lost grows as $1-e^2$, reaching zero only at e=1.

Solution 4 — The ballistic pendulum. Return →

(a) The impact lasts milliseconds; the external forces (gravity, string tension) are negligible over that interval, so momentum is conserved. During the impact the bullet deforms the block, converting kinetic energy into heat and deformation, so mechanical energy is **not** conserved. Afterwards, the block swings with no further non-conservative work, so energy is conserved again.

(b) Momentum: $mv = (m+M)V$. Energy: $(1)/(2)(m+M)V^2 = (m+M)gh$, so $V = \sqrt{(2gh)}$ and

$$v = (m+M)/(m)\sqrt{(2gh)}. \quad (3)$$

(c) $(K_f)/(K_i) = ((m+M)V^2)/(mv^2) = (m)/(m+M)$, so the fraction lost is $(M)/(m+M)$. With m=0.010 {kg} and M=2.0 {kg} this is 0.995 — over 99% of the energy is lost, which is why the device was abandoned in favour of electronic timing. **Check:** as $m\to\infty$ the fraction lost $\to 1$; as $m\to 0$ the bullet carries almost all the energy away in the block.

Solution 5 — Rolling race. Return →

(a) Energy with rolling constraint: Mgsin θ - f = Ma and fR = Iα with α = a/R. Eliminating f,

$$a = (g\sin\theta)/(1 + I/(MR^2)). \quad (4)$$

Hoop: 1+1 = 2, so a = 0.5gsin θ. Disc: 1+0.5 = 1.5, so a = 0.667gsin θ. Sphere: 1+0.4 = 1.4, so a = 0.714gsin θ.

(b) Times for the same distance scale as 1/√(a): t_({sphere}):t_({disc}):t_({hoop}) = 1: 1.035: 1.195. The sphere wins, the hoop loses.

(c) M and R cancel because both the driving force and the inertia scale with them; only the shape factor I/(MR²) survives. If the bodies slid without friction, a = gsin θ for all three and they would tie — friction is what breaks the symmetry. **Check:** the sphere's shape factor 2/5 is the smallest, giving the largest acceleration.

Solution 6 — Two blocks and a pulley. Return →

(a) a = (m_2-m_1)/(m_1+m_2)g = (2)/(8)(9.81) = 2.45 {m,s^(-2)} toward m_2's side. Tension: T = m_1(g+a) = 3.0(12.26) = 36.8 {N}.

(b) As m_2→ m_1, a→0 and T→ m_1g — static hanging. As m_1→0, a→ g and T→0: the heavy mass falls freely and the light mass provides no resistance.

(c) With a pulley of inertia I, the constraint becomes a = α R and the torque equation gives (T_2 - T_1)R = Iα. The result is

$$a = ((m_2-m_1)g)/(m_1+m_2+I/R^2), \quad (5)$$

so the pulley acts as an extra mass I/R². Setting I=0 recovers part (a). **Check:** the extra term is dimensionally a mass and positive, so it always slows the system.

Solution 7 — Explosion in mid-air. Return →

(a) 3mv = m(0) + 2mv' v' = 1.5v.

(b) K_i = (1)/(2)(3m)v² = 1.5mv²; K_f = (1)/(2)(2m)(2.25v²) = 2.25mv². Released energy = 0.75mv².

(c) The centre of mass continues on the original ballistic parabola: four seconds later it has advanced by 4v horizontally and dropped (1)/(2) g(4)² = 78.5 {m}. The fragments may fly in different directions, but the mass-weighted mean of their positions is exactly that point — which is what momentum conservation means when gravity is the only external force. **Check:** the released energy is positive, as it must be; the explosion adds kinetic energy but cannot change the total momentum.

Solution 8 — The falling chain. Return →

(a) When a length x has landed, the weight supported is (Mgx)/(L). The landing links stop from speed v = √(2gx), delivering momentum at rate (dp)/(dt) = (M)/(L)v· v = (M)/(L)(2gx) = (2Mgx)/(L). The scale reads the sum: (3Mgx)/(L). The factor 3 is one part weight plus two parts momentum flux.

(b) Once the last link has landed the reading drops to Mg — the dynamic term vanishes abruptly.

(c) A scale measures force, not mass: it reads the weight plus the rate of momentum transfer. **Check:** at x=L the instantaneous reading is 3Mg, so the needle briefly shows three times the weight before settling at Mg — a result students often disbelieve until they see it measured.

Solution 9 — Angular momentum and the skater. Return →

(a) No external torque about the vertical axis, so I_1ω_1 = I_2ω_2 ω_2 = 2.5ω_1.

(b) K_1 = (1)/(2) I_1ω_1², K_2 = (1)/(2)(0.4I_1)(6.25ω_1²) = 1.25K_1. The work done is +0.25K_1.

(c) Her muscles do that work by pulling mass inward against the centrifugal tendency. The reverse process — arms flying out while spinning faster — would require an energy source and a decrease in entropy; it cannot happen spontaneously, which is why skating spins are easy to speed up and hard to slow. **Check:** angular momentum rises by 0% and kinetic energy by 25%, so the energy must come from muscle work — it is not created from nothing.

Solution 10 — A variable-mass rocket. Return →

(a) In time dt the rocket of mass m ejects mass dm (with dm<0) at speed u relative to the rocket. Momentum conservation in the instantaneous rest frame of the rocket gives m,dv = -u,dm, so

$$\int_{v_0}^{v_f} dv = -u\int_{m_0}^{m_f}\frac{dm}{m} \quad \Delta v = u\ln\frac{m_0}{m_f}. \tag{6}$$

(b) With gravity, m,dv = -u,dm - mg,dt, so Δ v = uln (m_0/m_f) - gt.

(c) ln (m_0/m_f) = 7.8/3.0 = 2.6, so m_0/m_f = e^(2.6) = 13.5. Over 90% of the launch mass must be fuel — which is exactly why rockets are staged, and why a single-stage-to-orbit vehicle is so hard to build. **Check:** the ratio grows exponentially with the required Δ v/u, the central engineering fact of spaceflight.